NeuroVision AI - System Architecture & Engineering Specification

A deep-dive technical architecture document detailing model design, layer topologies, explainable AI mechanics, and data persistence.

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1. High-Level System Architecture

NeuroVision AI implements a clinical-grade medical computer vision pipeline designed for rapid diagnostic triage and full explainability:

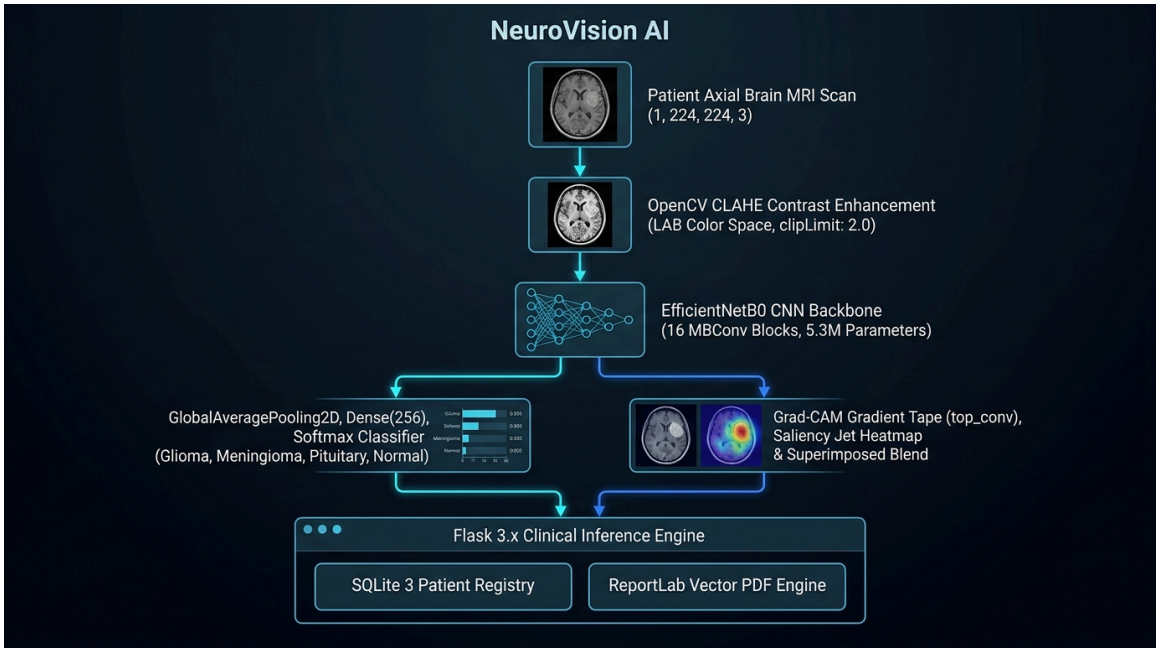
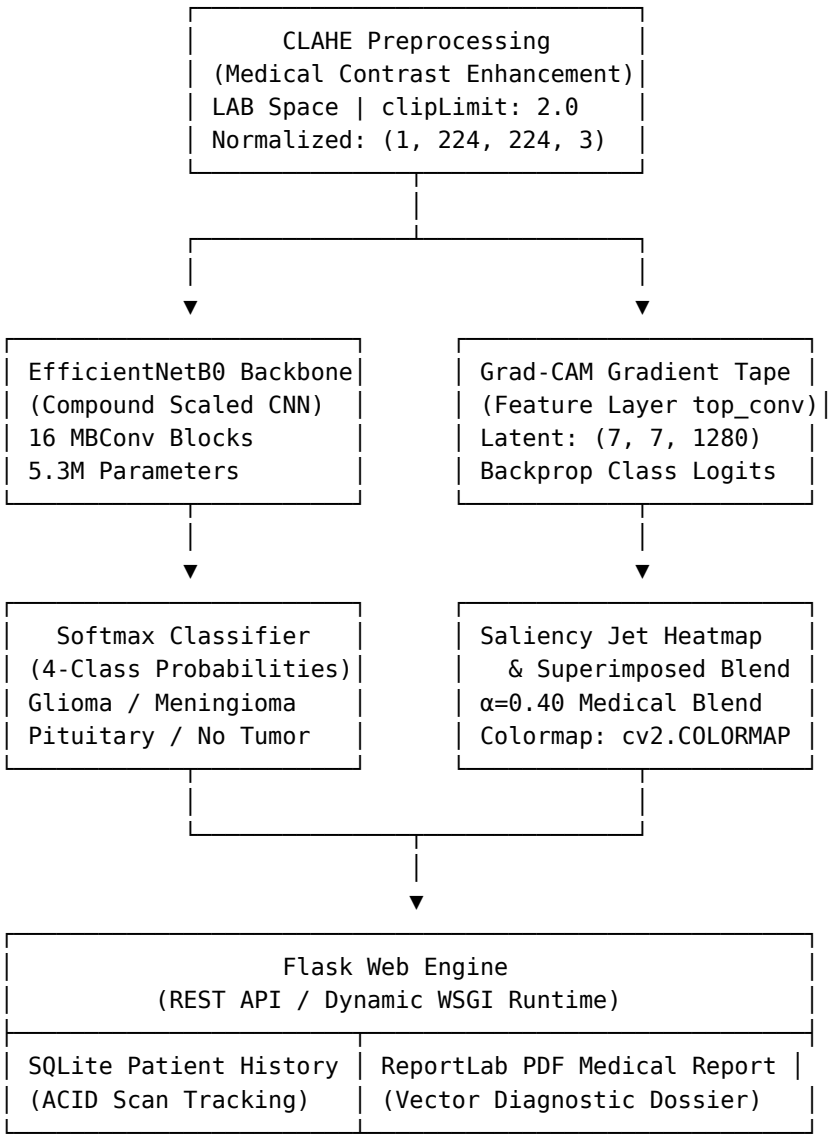


Figure 1: NeuroVision AI System Architecture

```
┌──────────────────────────────────┐
│      Patient Brain MRI           │
│   (Axial Scan / PNG / JPG)       │
│      Shape: (H x W x 3)          │
└──────────────────┬───────────────┘
                   │
                   ▼
```



1.1 Mermaid Pipeline Workflow

```
graph TD
    A[("Patient Brain MRI<br/>(Axial Scan / PNG / JPG / DICOM)")] --> B["CLAHE
Preprocessing<br/>(Contrast Limited Adaptive Histogram Eq.)<br/>LAB Space |
clipLimit=2.0 | tileGrid=8x8"]
    B --> C["Normalized Tensor<br/>Shape: (1, 224, 224, 3) | float32"]

    C --> D1["EfficientNetB0 Backbone<br/>(Compound Scaled CNN | 5.3M Params)"]
    C --> D2["Grad-CAM Gradient Tape<br/>(Feature Extraction Layer: top_conv)"]

    D1 --> E1["Global Average Pooling 2D<br/>Shape: (1, 1280)"]
    E1 --> F1["Dense 256 + Dropout 0.40<br/>L2 Regularization λ=1e-4"]
    F1 --> G1["Softmax Classifier<br/>(Glioma, Meningioma, Pituitary, No Tumor)"]
```

```
    D2 --> E2["Activation Map A^k: (7, 7, 1280)<br/>&part;y^c / &part;A^k
Gradient Backprop"]
    E2 --> F2["Neuron Importance Weights α_k^c<br/>ReLU Positive Attribution
Pooling"]
    F2 --> G2["Saliency Jet Heatmap<br/>Superimposed Medical Overlay (α=0.40)"]

    G1 --> H["Flask Web Engine & REST API Controller<br/>(/predict & /api/
predict)"]
    G2 --> H

    H --> I1[("SQLite Patient History<br/>neurovision.db | Indexed Search")]
    H --> I2["ReportLab Medical PDF Dossier<br/>Dual Imagery & Clinical
Protocols"]

    style A fill:#1e293b,stroke:#38bdf8,stroke-width:2px,color:#f8fafc
    style B fill:#1e293b,stroke:#818cf8,stroke-width:2px,color:#f8fafc
    style C fill:#0f172a,stroke:#94a3b8,stroke-width:1px,color:#f8fafc
    style D1 fill:#1e293b,stroke:#38bdf8,stroke-width:2px,color:#f8fafc
    style D2 fill:#1e293b,stroke:#f59e0b,stroke-width:2px,color:#f8fafc
    style G1 fill:#064e3b,stroke:#10b981,stroke-width:2px,color:#ecfdf5
    style G2 fill:#78350f,stroke:#f59e0b,stroke-width:2px,color:#fef3c7
    style H fill:#1e1b4b,stroke:#6366f1,stroke-width:2px,color:#e0e7ff
    style I1 fill:#1e293b,stroke:#10b981,stroke-width:2px,color:#f8fafc
    style I2 fill:#1e293b,stroke:#ef4444,stroke-width:2px,color:#f8fafc
```

---

2. Deep Learning Neural Architecture (EfficientNetB0)

2.1 Why EfficientNetB0?

Standard convolutional networks (VGG, ResNet) scale either depth, width, or resolution independently. EfficientNet balances all three dimensions uniformly via compound coefficient scaling:

$$\text{FLOPs} \approx 2^{\phi} \cdot \text{FLOPs}_{\text{baseline}}$$

For $\phi = 1$, EfficientNetB0 achieves high feature abstraction on medical neuroimaging with only **5.3 million parameters**, avoiding the severe overfitting risks common to models with 25M–138M parameters when trained on medical MRI datasets.

2.2 Layer-by-Layer Architectural Topology

```
Input Layer (Shape: 224 x 224 x 3, float32)
  |
  ▼
[Stem Conv2D: 32 filters, 3x3, Stride 2] + Batch Normalization + Swish
  |
  ▼
[Block 1: 1 MBConv Block, 16 channels, Expansion 1, Squeeze-and-Excitation]
  |
```

```
  ▼
[Block 2: 2 MBConv Blocks, 24 channels, Expansion 6, Stride 2, SE]
  │
  ▼
[Block 3: 2 MBConv Blocks, 40 channels, Expansion 6, Stride 2, SE]
  │
  ▼
[Block 4: 3 MBConv Blocks, 80 channels, Expansion 6, Stride 2, SE]
  │
  ▼
[Block 5: 3 MBConv Blocks, 112 channels, Expansion 6, SE]
  │
  ▼
[Block 6: 4 MBConv Blocks, 192 channels, Expansion 6, Stride 2, SE]
  │
  ▼
[Block 7: 1 MBConv Block, 320 channels, Expansion 6, SE]
  │
  ▼
[Final Conv2D: 'top_conv', 1280 filters, 1x1] <--- Grad-CAM Feature Tap
  │
  ▼
GlobalAveragePooling2D (Shape: 1280)
  │
  ▼
BatchNormalization (Channel-wise stability)
  │
  ▼
Dense Layer (256 units, ReLU, L2 Regularization λ=1e-4)
  │
  ▼
Dropout Layer (Rate: 0.40)
  │
  ▼
Dense Classification Output (4 units, Softmax Activation: Glioma, Meningioma,
Pituitary, No Tumor)
```

---

3. Two-Stage Transfer Learning Protocol

```
Phase 1: Feature Extraction (Epochs 1 - 10)
┌──────────────────────────────────────────┐
│ Backbone Weights Frozen (ImageNet)    │ ---> Only top custom head is trainable
│ Learning Rate: η = 1e-3 (Adam)        │      (GlobalAvgPool, Dense 256,
Softmax)
└──────────────────────────────────────────┘
                   │
                   ▼
Phase 2: Targeted Fine-Tuning (Epochs 11 - 25)
```

```
┌──────────────────────────────────────────┐
| Top 30 Backbone Layers Unfrozen        | ---> Fine-tunes high-level conv
features
| BatchNormalization Frozen              |      (Preserves running mean &
variance)
| Learning Rate: η = 1e-4 -> 1e-6        |      (Decays automatically on plateau)
└──────────────────────────────────────────┘
```

---

4. Grad-CAM Mathematical Formulation

Gradient-weighted Class Activation Mapping computes a visual explanation for the classification c:

1. **Gradient Computation:**

$$\frac{\partial y^c}{\partial A^k}$$

   Where y^c is the logits score for class c, and A^k is feature map k of layer `top_conv`.

2. **Neuron Importance Pooling:**

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{i,j}^k}$$

   Global average pooling captures the global importance of feature map k.

3. **Weighted Linear Combination & Rectification:**

$$L_{\text{Grad-CAM}}^c = \max\left(0, \sum_k \alpha_k^c A^k\right)$$

   The $\max(0, \cdot)$ ReLU function discards features that negatively impact class c, isolating the exact pixels that substantiate tumor presence.

---

5. Database Schema & Data Modeling

The platform utilizes a relational SQLite schema (`database/neurovision.db`):

```sql
CREATE TABLE predictions (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    patient_id TEXT NOT NULL,
    patient_name TEXT NOT NULL,
    patient_age INTEGER,
    patient_gender TEXT,
    physician_name TEXT,
    filename TEXT NOT NULL,
```

```
    gradcam_filename TEXT,
    predicted_class TEXT NOT NULL,
    confidence REAL NOT NULL,
    probabilities TEXT NOT NULL,  -- JSON serialized dictionary
    notes TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_predictions_class ON predictions(predicted_class);
CREATE INDEX idx_predictions_patient ON predictions(patient_name);
CREATE INDEX idx_predictions_created ON predictions(created_at);
```

---

6. Security, Privacy & HIPAA/GDPR Compliance

1. **Protected Health Information (PHI) Isolation:**
   - Patient records are contained inside a local SQLite database that never egresses outside the host network.
2. **Deterministic Hash Sanitization:**
   - Uploaded file names are stripped and renamed using UUID v4 and millisecond timestamps, preventing path traversal attacks.
3. **Payload Buffering Limits:**
   - Enforces a 16 MB ceiling on all incoming HTTP request bodies.
4. **Vector PDF Generation Security:**
   - PDF reports are compiled in-memory and streamed directly via `send_file`, ensuring zero exposure of uncompiled intermediary assets.